



ASSESSMENT 1 BRIEF

MODULE CODE	COM6033
MODULE TITLE	Applied Artificial Intelligence
MODULE LEADER	Dr Antesar Shabut
ASSESSMENT TITLE	Project (Software Artefact)
WEIGHTING	60%

ASSESSMENT LEARNING OUTCOMES

Upon successful completion of this assessment, you will be able to:

1. Apply the theoretical underpinnings of AI and machine learning algorithms and techniques to a specific problem.
2. Critically evaluate the principles and algorithms of AI and machine learning.
3. Evaluate the ethical considerations of AI, machine learning and computing.
4. Implement AI and machine learning algorithms to a specific problem.

INSTRUCTIONS

This project focuses on the design, implementation, and evaluation of AI and machine learning algorithms applied to a defined problem. You will train, test, and evaluate your model using a secondary dataset. If you plan to collect primary data, please discuss this with your module tutor and obtain ethical approval before proceeding. Your task is to formalise a problem statement, classify it as a classification, regression, or clustering problem, and then select, justify, and apply appropriate AI and machine learning algorithms and techniques to address it. This project is designed to assess your ability to apply theoretical knowledge, critically evaluate algorithmic choices, and reflect on the ethical considerations of AI-driven solutions. For the assessment, you are required to deliver the following:

- Select a dataset: Use trusted repositories like [Kaggle](#), [UCI Machine Learning Repository](#), [Google Dataset Search](#) or [OpenML](#) for a wide range of datasets.
- Define your problem: Clearly state the problem you want to solve using AI and machine learning.
- Clean the data: Prepare the dataset by handling missing values, duplicates, and normalisation.
- Design your solution: Choose the right algorithm(s) and build the model.
- Test and evaluate: Assess model performance with appropriate metrics like accuracy or MSE.
- Integrate into a web app: Deploy the model using Flask for user interaction.
- Reflect on the project: Summarise the design, challenges, and improvements for future work.

SUBMISSION DETAILS

RELEASE DATE	24 October 2025
SUBMISSION DATE	5 December 2025
DELIVERABLES	Your submission should include the following components: ○ Jupyter notebook: Ensure that your submission includes a Jupyter notebook with code documented and discussed. ○ Original or manipulated dataset: Include the dataset used, whether it is in its original form or has been modified for your analysis. ○ Zipped model deployment Code: Provide a zipped folder containing the code necessary for deploying your model. Evidence of completing all lab tasks: include proof that you have completed all the tasks outlined in the lab instructions (share your GitHub repository with the tutor).



SUBMISSION DETAILS

Submit your assignment by uploading it to Moodle before midday on the submission date. This deadline will be automatically and strictly enforced. If your submission is late, your grade may be affected. If you have any issues submitting your work, you must email the assessment team and copy in the module leader before the assessment due time. Do not leave your submission until the last minute to avoid any penalties due to problems with the submission portal.

Assessment Team: assessment@leedstrinity.ac.uk

Module Leader: a.shabut@leedstrinity.ac.uk

We may ask for a demonstration of your work following the submission. If needed, this will be communicated to you individually via email. Please check your emails regularly.

Your feedback / feed forward and mark for this assessment will be provided within 15 working days.

MARKING CRITERIA

Marks are awarded based on the following criteria. Within each part, aim to complete the work for each section before moving on to the next. The following banded marking scheme is used:

<i>Exceptional 1st</i>	100/95/92		<i>2:ii</i>	58/55/52
<i>Outstanding 1st</i>	88/85/82		<i>3rd</i>	48/45/42
<i>1st</i>	78/75/72		<i>Bare Fail</i>	38/35/32
<i>2:i</i>	68/65/62		<i>Fail</i>	25/20/10/0

If you have completed all the preparatory exercises and attended your classes, the estimated additional time required to PASS this assessment is approximately 60 hours.

To obtain a 3 rd mark (40%), you must have:	- Shown limited understanding and applied it to the problem. - Demonstrated a basic application of data exploration, manipulation, and cleaning techniques. - Applied a basic set of machine learning and coding skills and produced a mostly accurate model. - Included limited or no use of documentation, discussion, and reflection throughout the project. - Carried out a basic analysis but remains descriptive with few or no insights or conclusions.
To obtain a 2:ii mark (50%), you must have (in addition to the above):	- Shown some understanding and applied it to the problem. - Demonstrated some application of data exploration, manipulation, and cleaning techniques. - Applied a good set of machine learning and coding skills and produced a mostly accurate model. - Included a consistent discussion and reflection on the project and good documented code. - Offered a good solution to the problem with some evidence to justify the selection of techniques. - Carried out an analysis throughout with some general conclusions drawn.
To obtain a 2:i mark (60%), you must have (in addition to the above):	- Shown a good understanding and applied it well to the problem. - Included a logical solution to the problem with some level of creativity in forming the problem. - Demonstrated a good application of data exploration, manipulation, and cleaning techniques with mostly accurate model and some coherent and working parts of the artefact.



To obtain a 1st mark (70%), you must have (in addition to the above):	○ Applied a consistently good set of machine learning and coding skills. ○ Included consistent and accurate discussion and reflection on the project and well documented code. ○ Offered a logical solution to the problem with well selected evidence to justify the selection of techniques. ○ Carried out a strong analysis throughout with some appropriate conclusions drawn.
	○ Shown a thorough understanding and applied it very well to the problem. ○ Included a clear and logical solution to the problem with a very good level of creativity in forming and solving the problem. ○ Demonstrated a very good application of data exploration, manipulation, and cleaning techniques with accurate model and mostly coherent and working artefact. ○ Applied a very good and comprehensive set of machine learning and coding skills. ○ Included excellent discussion and reflection on the project and well documented code. ○ Carried out insightful analysis throughout with appropriate conclusions drawn.

USE OF GENERATIVE ARTIFICIAL INTELLIGENCE IN THIS MODULE

You may use generative AI such as Microsoft Copilot to assist you in the process of undertaking the assessment in the following ways: brainstorming, research, planning, feedback, editing.

All use of generative AI must be explicitly acknowledged, and any artificially generated content (e.g. images) explicitly labelled, with the source of the AI tool referenced using current APA referencing conventions. You can find further guidance on the library website on their AI webpage).

In submitting your assignment, you agree to disclose the extent to which you have used generative AI in preparing this work and include evidence of your AI use in your appendices (e.g. dated screen shots of your use of this tool or copy and paste your AI chat into Word).

Failure to disclose your generative AI use may result in a zero for your assignment and a referral for academic misconduct (see the Student Academic Misconduct Policy under Essential Info in the MyLTU app).

Include one of the following statements on your assignments.

Either:

This assignment used generative AI in the following ways for the purposes of completing the assignment (choose 1 to 5 of the following): brainstorming, research, planning, feedback, editing.

Or:

This assignment did not use generative AI for the purposes of completing the assignment.

ACADEMIC MISCONDUCT

Academic Misconduct includes all forms of academic dishonesty, whether intentional or accidental, that compromise the integrity of the University's assessment processes. It is essential that you review our [Student Academic Misconduct Policy](#) to understand the guidelines and the serious consequences that may arise if they are not followed.



HELP AND SUPPORT

- Please use the module handbook and the [Computer Science Community Teams site](#) as a source of information. Do try and find the answer out yourself before reaching out for help.
- Support will be provided via Microsoft Teams and email during office working hours. You can also ask questions during your timetabled sessions. You may request a one-to-one meeting with a tutor during their published office hours.
- The Student Support team are available seven days a week to support you in all aspects of student life. This could be for support relating to your course, your accommodation or for more general advice such as relationships or your wellbeing. Log in to the LTU app to access support services.
- The full set of university guidelines on assessments, deadlines, and extensions is available on the LTU app, please familiarise yourself with the documentation.